

Standard Secondary Biology Curriculum

Subject Code: BIO-10 | Official Academic Curriculum

Verified learning outcomes and required chapter competencies for students.

Chapter 1: Life Processes

- Autotrophic and heterotrophic nutrition
- Photosynthesis mechanism
- Aerobic and anaerobic respiration
- Human circulatory system and blood
- Excretion in humans and nephron function

Chapter 2: Control and Coordination

- Nervous system and reflex arc
- Human brain structure and functions
- Plant hormones and tropism movements
- Animal endocrine glands and hormones

Chapter 3: How do Organisms Reproduce

- Asexual reproduction binary fission budding
- Sexual reproduction in flowering plants
- Male and female human reproductive systems
- Reproductive health and contraception

Chapter 4: Heredity and Evolution

- Mendel laws of inheritance monohybrid dihybrid
- Sex determination in human beings
- Speciation and evolutionary relationships

Chapter 5: Our Environment & Ecosystems

- Food chains and food webs
- Trophic levels and energy flow 10 percent rule
- Ozone layer depletion and waste disposal